



October University for Modern Sciences and Art

Faculty of Computer Science

Graduation Project

SENTINEL (Smart Elderly Navigation & Telehealth Integrated Lifeline)

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Preliminary Pages

Declaration / Acknowledgment

We do hereby declare that this graduation project titled “SENTINEL (Smart Elderly Navigation & Telehealth Integrated Lifeline)” has been prepared by us alone and this project has been carried out under the guidance of the assigned supervisor of this project. The project has never been used in its entirety or partially for any other academic qualification.

All the sources from which information has been taken for carrying out this project have been properly cited and referenced.

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Abstract

Hypertension is a top risk factor causing heart-related problems, especially in older people, and thus, constant BP monitoring is necessary. Cuff-less methods of BP estimation based on physiological signals and artificial intelligence have been developed because of inconvenience caused by traditional BP measuring devices to constantly measure BP.

This thesis introduces SENTINEL (Smart Elderly Navigation & Telehealth Integrated Lifeline), an intelligent health monitoring platform which utilizes deep learning-based cuff-less blood pressure estimation and a web-based monitoring system designed specifically for elderly patients and their families. In this framework, systolic and diastolic blood pressures will be estimated in parallel by using multi-task learning techniques.

In this thesis, we examine three physiological signals (ECG, PPG and RESP) in combination with various deep learning architectures (CNN, RNN, LSTM, GRU, Bi-LSTM) to find optimal combinations of modality and architecture. Based on these results, multimodal learning will be tested for dual-modalities and tri-modalities in conjunction with four different fusion techniques: feature concatenation, bidirectional cross-attention, Gated Multimodal Unit (GMU), and a proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF).

The experimental validation on the public MIMIC-BP dataset showed that PPG is the best modality in terms of single-modality prediction. It was proved that the proposed DGAGF framework efficiently combines complementary information coming from other modalities. In addition, the combination of PPG and RESP turned out to be the best multimodal combination and outperformed a conventional combination of ECG and PPG. It was shown that proper fusion methods have a stronger influence on prediction results than the increase in the number of used modalities.

In order to make the proposed framework more applicable, the trained AI models were incorporated in the designed monitoring application implemented with the help of React, FastAPI, and SQLite, allowing for patient management, prediction history, alerts generation, and remote monitoring of relatives.

Keywords

Cuffless Blood Pressure Estimation, Deep Learning, ECG, PPG, Multimodal Learning,
Physiological Signals

Chapter 1: Introduction

1.1 Background

The increase in the number of elderly people together with the rise in cases of cardiovascular diseases has led to the emergence of the need for intelligent monitoring systems in healthcare. Among physiological parameters, the Blood Pressure (BP) plays a critical role for assessing the health condition of the cardiovascular system, diagnosing hypertension, and providing medical intervention. Traditionally, BP monitoring systems make use of cuffs, which perform intermittent measurements and might be uncomfortable for regular and long-term monitoring.

Recent technological advancements in terms of physiological sensing, physiological signals processing, artificial intelligence, and deep learning allow estimating blood pressure without a cuff based on physiological parameters including Electrocardiogram (ECG), Photoplethysmography (PPG), and Respiration (RESP). Such solutions are able to provide continuous, non-invasive monitoring of the cardiovascular system while improving patients' comfort and providing remote healthcare services.

SENTINEL (Smart Elderly Navigation & Telehealth Integrated Lifeline) represents an intelligent monitoring platform for elderly people's health care which incorporates the proposed intelligent cuffless blood pressure estimation approach based on deep learning along with the web-based platform for remote monitoring of patients. The research studies various physiological modalities, different deep learning architectures, multimodal fusion methods, and introduces a new multimodal fusion framework called Dominant-Guided Auxiliary Gated Fusion (DGAGF). Moreover, the developed web platform allows uploading physiological signals, obtaining blood pressure estimation, and providing the opportunity for relatives to monitor the patient's history and health status.

1.2 Problem Statement

High blood pressure is considered one of the major factors of cardiovascular disease, especially in the case of elderly patients who are required to constantly monitor their blood pressure.

Despite the number of research on cuffless blood pressure prediction using artificial intelligence algorithms, there still lacks the use of such technologies in an intelligent healthcare monitoring framework. Moreover, multimodal frameworks usually give equal weight to each of the physiological signals used for fusion, while in practice, different modes influence the blood pressure prediction in different ways.

This is why there is a need for an intelligent healthcare monitoring framework which would combine high-quality cuffless blood pressure prediction, multimodal fusion, assessment of the health condition of the patient, generation of alerts, monitoring, and supervising of patients remotely.

1.3 Motivation

The rationale behind this study is that it will enhance the health care services of older people through constant, comfortable, and smart blood pressure monitoring. The frequent use of cuffs may not be very convenient and suitable for extended use, but the AI-enabled estimation without the use of a cuff is a viable solution.

Furthermore, the integration of artificial intelligence with web technology helps the relatives to remotely monitor their elderly family members, thus helping in making them aware of their health issues and responding quickly to their abnormal blood pressure levels.

1.4 Objectives

The main goal of the project will be to design an intelligent and innovative cuffless BP monitoring system for elderly health care by utilizing deep learning techniques and designing a monitoring web application platform.

Objectives of the Project include:

Design of precise AI models for estimating SBP and DBP from ECG, PPG, and RESP signals in a cuffless fashion.

Exploration of single modality as well as multimodal learning techniques, comparison of various deep learning algorithms, and analysis of different fusion schemes, including the newly proposed DGAGF method.

Implementation of an advanced integrated web platform for uploading physiological signals, predicting blood pressure, visualizing health condition, generating alerts, keeping monitoring history, and remote patient monitoring by family members.

1.5 Scope

The research prototype to be developed for this project includes two main components: AI-enabled blood pressure estimation without a cuff and the web-enabled healthcare monitoring system.

The current implementation of the developed platform involves patient/relative registration, patient authentication, physiological data upload, blood pressure prediction, health condition visualization, alerting, monitoring history, report generation, and database management.

However, at the moment, there is no implementation involving integration with wearables, real-time streaming of physiological signals, hospital information systems, cloud computing, and clinical diagnosis. The above-listed features represent further work for the system under development.

1.6 Expected Contribution

This research brings about a contribution to both the artificial intelligence field and software engineering.

On the aspect of AI, the research is focused on benchmarking several physiologic modalities including ECG, PPG, and RESP, evaluating various architectures of deep learning, exploring multimodal fusion methods, and introducing Dominant-Guided Auxiliary Gated Fusion (DGAGF) method for estimating cuffless blood pressure.

On the aspect of software engineering, the research focuses on building a health monitoring system that combines AI prediction algorithms with web application development, database management, patient relative interactions, and software testing.

1.7 SDG Relevance

In addition to contributing to UN SDG 9 (Industry, Innovation, and Infrastructure), which involves the application of artificial intelligence, deep learning, and advanced software technology in healthcare, this project specifically helps with the achievement of UN SDG 3 (Good Health and Well-being), which involves continuous blood pressure monitoring, early risk detection, preventative medicine, and better accessibility for remote health monitoring.

1.8 Thesis Organization

The rest of this dissertation is structured as follows.

Chapter 2 provides an introduction to literature review, background, related work, physiological signals, deep learning approaches, multimodal fusion techniques, and health monitoring systems.

Chapter 3 introduces the research methodology, data set, data preprocessing procedure, development of AI model, the proposed DGAGF framework, experiment design, and software prerequisites.

Chapter 4 describes the implementation of the AI system and web application for monitoring purposes, consisting of software architecture, UML models, database schema, APIs, and technologies used.

Chapter 5 describes the experimental results and performance assessment, software testing, user experience assessment, and validation of the system.

Finally, **Chapter 6** summarizes the main findings, discusses limitations, and outlines future research directions.

Chapter 2: Literature Review / Background

2.1 Related Work

The latest advancements in artificial intelligence (AI) and physiological signal processing have made the development of cuffless blood pressure estimation devices very fast. In contrast to conventional devices, current research efforts focus on estimating systolic blood pressure (SBP) and diastolic blood pressure (DBP) based on physiological signals such as electrocardiogram (ECG), photoplethysmography (PPG), and respiration (RESPIRATION). Deep learning models exhibit good results when automatically extracting useful features from the physiological signals, hence the need for feature engineering is diminished [1], [2], [3].

One of the first multimodal deep learning frameworks has been proposed by Liang et al. [5] and it concerns hypertension prediction based on ECG and PPG signals from the MIMIC database. They showed that combination of cardiovascular signals results in higher informativity of features comparing to a single modality case. However, the framework was designed specifically for hypertension classification task, not for continuous blood pressure estimation and advanced methods of multimodal fusion have not been considered.

In [9] Xu et al. suggested a deep learning framework for estimation of blood pressure using synchronized ECG and PPG signals. It has been shown that integration of both cardiovascular signals yields better results comparing to the single modality case. However, the framework evaluates only two physiological signals, while other complementary signals, such as respiration have not been investigated. Moreover, only one deep learning framework has been analyzed, the comparisons of different architectures have not been performed.

Considering single modality prediction problem, Koparır and Arslan [10] proposed CNN framework for cuffless blood pressure estimation based on PPG signals only. They showed that CNNs are able to automatically extract morphological features from the raw PPG waveform and provide competitive prediction performance without feature extraction. However, only one physiological signal was considered in their paper, and the question whether multimodal learning helps improve the performance remains unanswered.

More recently, Haque et al. [4] have designed an ensemble based artificial intelligence approach to estimate the continuous blood pressure. Their work uses an ensemble of machine learning models to enhance prediction accuracy and proved that ensemble learning can minimize estimation errors when compared to individual models. However, despite these achievements, the suggested model did not consider any aspect of deploying such a system into healthcare applications, patient monitoring, and interaction with users.

The development of attention-based deep learning models has also affected the research on cuffless blood pressure estimation. Fan et al. [11] have proposed the use of Transformer architecture that merges Conformer and Swin Transformer architectures to analyze the physiological signal. The authors have shown a very high prediction accuracy using long-range dependencies in physiological signals. However, Transformer architectures require much more computational resources and data than traditional CNNs and RNNs.

Understanding the need for combining various physiological cues, Hassan et al. [12] have introduced MuFuBP-Net, a multimodal fusion network, which fuses features of ECG, PPG, and RESP signals through feature fusion using deep learning techniques. The results obtained by the authors proved the efficacy of multimodal learning approaches when it comes to improving the prediction accuracy in comparison with single-modality solutions. Nevertheless, the method proposed by the authors assumes that all the physiological cues have the same importance when it comes to fusing features.

Recently, several attempts were undertaken for personalized BP estimation. Li et al. [15] developed a few-shot personalized learning approach, which allows adapting the predictive model according to particular physiological properties of an individual user. Even though personalization leads to improvement in prediction accuracy, the approach needs further subject-specific adaptation.

Another recent research paper published in [16] by Suleimenova et al. proposed the PiSO optimization methodology which could be used to choose suitable pre-processing methods and deep learning models for estimating arterial blood pressure with the help of the MIMIC-IV database. Their main objective was the importance of both the optimization of preprocessing methods and deep learning models. However, the focus of their research is mainly on the optimization techniques and not on the multimodal physiological fusion or healthcare systems integration.

Apart from prediction techniques, some researchers have studied intelligent health monitoring systems. Mendoza Ramos et al. [29] proposed a healthcare monitoring platform based on IoT technologies in combination with deep learning in remote cardiovascular monitoring. It showed that the application of artificial intelligence in healthcare systems is possible. The suggested platform was not designed specifically for the cuffless blood pressure estimation via multiple physiological signals.

Generally speaking, past research works show that there have been remarkable advancements in estimating the blood pressure without using cuff methods through employing artificial intelligence techniques and physiological signals. However, current research works tend to emphasize more on improving the estimation accuracy of cuffless blood pressure but pay little attention to comprehensive benchmarking on various physiological signals, comparison of deep neural network structures, multimodal adaptive fusion approach, and building a healthcare monitoring platform.

2.2 Comparison of Existing Approaches

Table 2.1 summarizes the most relevant studies related to cuffless blood pressure estimation. The comparison considers the physiological modalities, datasets, deep learning models, evaluation methodologies, and limitations of each study. Since different studies employ different datasets, preprocessing pipelines, and evaluation protocols, their reported performances should be interpreted carefully rather than being considered directly comparable.

Study	Dataset	Physiological Modalities	AI Model	Evaluation	Main Contribution	Limitations	Comparability
Liang et al. [5]	MIMIC	ECG + PPG	Deep Learning	Hypertension Assessment	Demonstrated the benefit of combining ECG and PPG signals	Focused on hypertension classification rather than BP estimation	Partial comparison due to different prediction target
Xu et al. [9]	Physiological Signals Dataset	ECG + PPG	Deep Learning	BP Estimation	Improved prediction accuracy using multimodal learning	Did not evaluate RESP or compare multiple architectures	Different dataset and evaluation protocol
Koparır and Arslan [10]	PPG Dataset	PPG	CNN	BP Estimation	Demonstrated effective feature extraction using CNN	Single-modality approach only	Different dataset and preprocessing
Haque et al. [4]	Health Monitoring Dataset	Physiological Signals	Ensemble Learning	Continuous BP Estimation	Improved robustness using ensemble models	No healthcare monitoring platform	Different dataset and experimental setup
Fan et al. [11]	PPG Dataset	PPG	Conformer + Swin Transformer	BP Estimation	Captured long-range dependencies using Transformer models	High computational complexity	Different model architecture and dataset
Hassan et al. [12]	MIMIC-Based Dataset	ECG + PPG + RESP	MuFuBP-Net	BP Estimation	Demonstrated the effectiveness of multimodal fusion	Equal treatment of all modalities and research prototype only	Similar research objective but different fusion strategy

Study	Dataset	Physiological Modalities	AI Model	Evaluation	Main Contribution	Limitations	Comparability
<i>Li et al. [15]</i>	<i>Personalized Physiological Dataset</i>	<i>PPG</i>	<i>Personalized Deep Learning</i>	<i>Personalized BP Estimation</i>	<i>Improved subject-specific prediction accuracy</i>	<i>Requires personalized adaptation for each subject</i>	<i>Different objective and evaluation protocol</i>
Proposed SENTINEL System	MIMIC-BP	ECG + PPG + RESP	CNN, RNN, LSTM, GRU, BiLSTM + DGAGF	MAE, MSE	Comprehensive benchmarking of physiological modalities, evaluation of multiple deep learning architectures, adaptive multimodal fusion through DGAGF, and deployment within an end-to-end healthcare monitoring platform	Real-time wearable integration and large-scale clinical validation remain future work	Benchmark performed using identical preprocessing, dataset split, and evaluation protocol across all experiments

The comparison made in Table 2.1 clearly indicates the extent to which prior research differs in their use of datasets, physiological modalities, pre-processing methods, deep learning models, and assessment criteria. Therefore, direct comparisons through numbers cannot be done in such cases. In contrast, prior research can be seen as complementary in that they demonstrate the efficiency of various approaches under varying conditions.

Different from prior research, the proposed SENTINEL system conducts a fair comparison by assessing all candidate models under the same conditions of data (MIMIC-BP), pre-processing pipeline, subject-wise train/ validation/test split, and evaluation measures.

2.3 Research Gap

Based on the findings from the previous section, the following is clear: great progress has been made in estimating the cuffless blood pressure via the use of machine learning techniques, in particular, deep learning techniques. Indeed, numerous researchers have been able to leverage deep learning for the estimation of systolic and diastolic blood pressure via the use of ECG, PPG, or physiological multimodal signals. However, there are some gaps in the research conducted on this topic that should be addressed.

One such gap is that the majority of studies evaluate the efficacy of only a small set of physiological modalities. While most of these studies use PPG signals alone, others use PPG and ECG but do not investigate whether respiration data is of any value in estimating blood pressure. As a result, we do not have enough information about which type of physiological signal gives the most informative representation when evaluating it in the same experimental setting.

Another gap is that previous studies evaluate only one or two neural network architectures. Given that different signals have different characteristics, it is not reasonable to assume that one neural network will give the best results for all kinds of physiological signals.

Third, even though the multimodal learning paradigm has received considerable attention from the community, most existing fusing methods give equal significance to all physiological modalities in the process of feature fusion. Nevertheless, past literature shows that all physiological modalities do not contribute equally to blood pressure predictions. More specifically, PPG usually contributes more predictive information than ECG and respiratory signals. Thus, giving equal importance to all physiological modalities might add unnecessary noise to the learnt representation and degrade prediction results.

Also, while prediction accuracy receives significant consideration from the majority of past works, practical deployment is usually overlooked. Patient registration, history of predictions, alert visualization, remote monitoring, and relative supervision are some of the health care features that have rarely been included in any complete health care monitoring system.

In order to overcome such shortcomings, the presented SENTINEL system explores the issue of cuffless blood pressure estimation from the perspectives of artificial intelligence and software engineering. In this regard, the research conducts a thorough comparison of three physiological modalities (ECG, PPG, and RESP) with respect to five deep learning algorithms (CNN, RNN, LSTM, GRU, and BiLSTM), applying identical pre-processing steps and data splitting on subject level. According to the results obtained, multimodal learning is explored in the form of two- and three-modality combinations with various approaches to signal fusion. Additionally, this paper suggests Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework that allows one to exploit complementary physiological signals based on identification of dominant and auxiliary modalities during the process of feature fusion. Moreover, the developed AI models are integrated in the SENTINEL healthcare monitoring system that allows for blood pressure prediction, patient management, alerts, history monitoring, and relatives' remote supervision in a single web-based platform.

This work therefore provides a comprehensive benchmark for cuffless blood pressure estimation and suggests the DGAGF framework for multimodal signal fusion.

2.4 Theoretical Background

In order to discuss the principles of the designed SENTINEL system, certain theoretical terms that belong to the biomedical engineering, artificial intelligence, signal processing of physiological data and healthcare monitoring fields should be introduced. Comprehension of these terms will provide an insight into the logic behind the cuffless BP measurement approach and design solutions applied during this research.

Blood pressure is one of the physiological parameters that help estimate cardiovascular state of patients. It is defined as the arterial pressure created by blood flow on the arterial walls. From the medical perspective, the term blood pressure includes two measures: systolic blood pressure (SBP), the maximal arterial pressure achieved during the process of ventricular contraction, and diastolic blood pressure (DBP), the minimal arterial pressure reached when the heart is resting after each contraction. Continuous monitoring of the two above-mentioned parameters is crucial for detecting hypertension and other cardiovascular diseases especially in elderly patients who are prone to developing cardiovascular disorders [2], [28].

However, conventional blood pressure measurement methods depend on cuff-based sphygmomanometers which use temporary blood flow obstruction to determine the pressure in arteries. While such measurement techniques offer accurate clinical results, their use in continuous monitoring of blood pressure is not recommended due to their interference with patient's normal activities and pain associated with repetitive inflation of the cuffs [1], [2]. These facts have prompted researchers to explore alternative approaches for blood pressure determination based on physiological parameters and Artificial Intelligence technologies [1], [2].

In previous cuffless BP estimation studies, Electrocardiogram (ECG), Photoplethysmography (PPG), and Respiration (RESP) have shown themselves to be the most valuable physiological signals for the determination of blood pressure. ECG measures the electrical activity of the heart providing important data about heart rhythm and electrical conduction. Due to its close relation with blood pressure, ECG has been widely used in blood pressure measurement studies. However, ECG is only concerned with electrical activity of the heart but not its vascular function making it less useful as an independent predictor of blood pressure [5], [8].

Photoplethysmography (PPG) is an optical method of measuring variations in the peripheral blood volume by means of the infrared or visible light. The PPG waveform shape contains many valuable vascular properties such as pulse amplitudes, waveform shape, systolic peak, and pulse width, all of which are closely associated with arterial stiffness and blood pressure variations. Because of its simplicity, low cost, and suitability for wearing device integration, PPG has gained popularity as the most popular physiological signal for estimating the blood pressure without the use of the cuff. Some recent papers confirmed that deep learning models, specifically Convolutional Neural Network models (CNN), were capable of extracting useful representations directly from the raw PPG waveform without any need for hand-crafted features [3], [10], [13].

The RESPIRATION (RESP) signal offers complementary physiological information on breathing activity and dynamics of the process. Variations of the intrathoracic pressure during breathing affect venous return, cardiac output, heart rate, and arterial pressure; therefore, respiration indirectly modulates blood pressure. Though respiration was not considered as widely as the ECG and PPG in the previous papers on blood pressure estimation, recent physiological research proved that it provides complementary information that could help multimodal models [24]–[27].

Advances in artificial intelligence have greatly contributed to improving physiological signal processing capabilities. Traditional machine learning approaches require manually created features for analysis purposes. In turn, deep learning systems learn hierarchical feature representations automatically from input data without any pre-processing. CNN architecture is specifically dedicated to recognizing local morphology of input signals, therefore it is the best choice for PPG wave analysis. RNN and its modifications such as LSTM and GRU architectures are especially good at modeling temporal dependencies in physiological sequences. BiLSTM network architecture takes advantage of bidirectional processing of physiological sequences [2], [4], [12].

Although single-modality models have demonstrated promising prediction performance, recent research has increasingly shifted toward multimodal learning. Since ECG, PPG, and RESP capture different physiological characteristics, combining them allows complementary information to be exploited while reducing the limitations associated with individual modalities. Various multimodal fusion techniques have therefore been proposed, including feature concatenation, attention-based fusion, and gated fusion mechanisms. Feature concatenation simply combines feature vectors extracted from multiple modalities, whereas attention mechanisms dynamically learn relationships between physiological signals. Gated fusion approaches, such as the Gated Multimodal Unit (GMU), further improve multimodal learning by adaptively controlling the contribution of each modality during feature integration [11], [12].

In spite of these advancements, current fusion methods consider all the physiological modalities to have the same contributions towards estimating blood pressure. On the contrary, empirical results found in previous literature indicate that various physiological features exhibit varying abilities to predict. For example, PPG often has more ability to estimate blood pressure than ECG and respiration [3], [10]. As a result of such observations, in this paper, a Dominant-Guided Auxiliary Gated Fusion (DGAGF) model is introduced, which takes into account dominant and auxiliary physiological modalities in multimodal learning. In the case of DGAGF, all physiological modalities are not considered to have the same contribution. Only in those cases where auxiliary modalities improve the prediction ability do they become part of the framework.

Last but not least, the proposed AI model is incorporated into the overall healthcare monitoring system designed using modern software engineering techniques. The system is built based on a client-server design pattern in which the frontend is responsible for the user interface of the patients and their relatives whereas the backend executes the processing of physiological signals, blood pressure prediction, database handling, and alerts generation using the RESTful API technology.

2.5 Datasets & Technologies

The design and validation of the proposed SENTINEL system are based on the use of open physiological datasets and software solutions. The above-mentioned resources serve as a basis for designing, training, validation, and implementation of the suggested cuffless blood pressure estimation approach.

The experimental analysis carried out within the framework of the present study is based on the usage of the MIMIC-BP dataset [20] as a public benchmark dedicated to the problem of cuffless blood pressure estimation. The dataset comprises synchronized physiological data along with blood pressure measurements, which can be used for performing a supervised learning task for the estimation of systolic blood pressure (SBP) and diastolic blood pressure (DBP). Each of the recordings involves Electrocardiogram (ECG), Photoplethysmography (PPG), and Respiration (RESP) signals synchronized with arterial blood pressure values.

The MIMIC-BP dataset was chosen due to some of its benefits in contrast to other public datasets. Firstly, it contains more than one physiologic modality synchronized at once, which makes it possible to conduct comprehensive multimodal learning experiments. Secondly, it consists of physiologic data preprocessed in a standardized way and of a high quality appropriate for deep learning tasks. Lastly, the dataset is used more and more frequently as a benchmark for recent studies on cuffless blood pressure estimation [20].

Several deep learning architectures were examined in this study to choose the most appropriate architecture for each physiologic modality. Deep learning architectures such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU) and Bidirectional Long Short-

Term Memory (BiLSTM) were examined in this study. These deep learning architectures were chosen because of their wide adoption in physiologic signals analysis and their successful usage in recent studies on cuffless blood pressure estimation [4], [10]–[12].

As the goal of the research is to evaluate the impact of using more than one physiological modality on prediction accuracy, different approaches to multimodal fusion will be evaluated in this work in addition to the aforementioned single-modality models. Apart from traditional feature concatenation, attention-based fusion, and GMU approaches, the present research proposes to use Dominant-Guided Auxiliary Gated Fusion (DGAGF).

The implementation of the SENTINEL platform was done using state-of-the-art software development technologies, including those that allow the integration of artificial intelligence models into a healthcare monitoring application. The frontend of the platform was written using the React.js framework, which makes it possible to create responsive and interactive user interfaces for patients and their relatives. The backend was built using the FastAPI framework, providing RESTful API endpoints for user authentication, physiological signal processing, blood pressure prediction, monitoring history fetching, and alerts management.

Deep learning models were created using TensorFlow and Keras libraries, which make the process of model building easy. Prediction results, user accounts, monitoring history, and alerts are saved into the SQLite database management system. Interaction between frontend and backend components is done via HTTP REST APIs.

In order to achieve quality software, the API endpoints were validated using Postman, while the back end was tested using unit tests and integration tests. The whole implementation process was done with the help of Visual Studio Code as the main integrated development environment. All these technologies allowed for building an end-to-end healthcare monitoring system based on artificial intelligence and software engineering.

Chapter 3: Methodology and System Design

3.1 Research Methodology

This project follows an experimental research methodology to develop and evaluate a cuffless blood pressure monitoring system using physiological signals and deep learning techniques. The selected methodology is appropriate for the problem because it enables systematic investigation, quantitative evaluation, and comparison of different physiological modalities and machine learning approaches for blood pressure estimation.

The research process was conducted through a series of structured stages. Initially, physiological signals and blood pressure measurements were obtained from the MIMIC-BP dataset. The collected data were then prepared through a preprocessing stage to ensure consistency and suitability for model development.

Subsequently, multiple deep learning architectures were implemented and evaluated to investigate their effectiveness in estimating systolic blood pressure (SBP) and diastolic blood pressure (DBP). The study considered both single-modality and multimodal learning approaches in order to assess the contribution of individual physiological signals as well as combined modalities.

The developed models were evaluated using standard regression metrics under controlled experimental conditions. Subject-level dataset splitting was adopted throughout the experiments to ensure fair evaluation and prevent data leakage between training, validation, and testing subsets.

Finally, the best-performing prediction models were integrated into the SENTINEL monitoring platform to provide an accessible environment for blood pressure prediction, monitoring, and result visualization.

3.2 Proposed Approach

The proposed SENTINEL framework aims to provide a non-invasive and AI-powered solution for cuffless blood pressure monitoring using physiological signals. The framework combines physiological signal processing, deep learning-based prediction, and a monitoring platform to estimate systolic blood pressure (SBP) and diastolic blood pressure (DBP) without the need for traditional cuff-based devices.

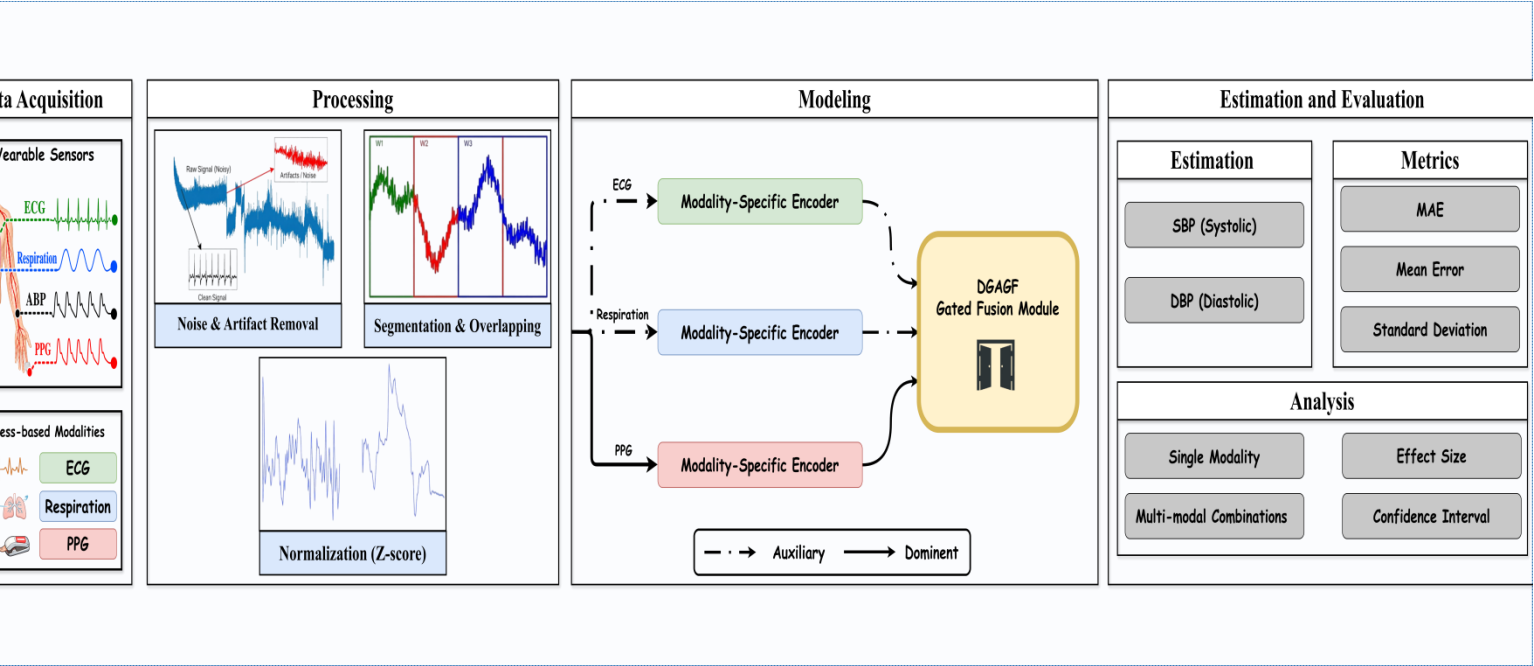
The proposed approach consists of five major stages: data acquisition, preprocessing, feature learning and prediction, multimodal fusion, and system deployment. Initially, synchronized ECG, PPG, and RESP signals are obtained from the MIMIC-BP dataset and prepared for analysis through a standardized preprocessing pipeline.

The processed signals are then supplied to deep learning models that learn the relationship between physiological patterns and blood pressure values. A multi-task learning framework is employed to jointly estimate SBP and DBP from the extracted signal representations.

To investigate the contribution of different physiological modalities, the framework evaluates both single-modality and multimodal learning strategies. In the multimodal stage, information from multiple physiological signals is combined using advanced fusion mechanisms, including the proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework, to improve prediction performance and exploit complementary physiological information.

Finally, the trained prediction models are integrated into the SENTINEL monitoring platform, enabling users to upload physiological signals, obtain blood pressure predictions, monitor historical records, and receive health status feedback through an accessible web-based interface.

3.3 System Architecture



3.4 Algorithms / Models

This project employs several deep learning architectures to investigate their effectiveness in cuffless blood pressure estimation using physiological signals. The selected models represent both convolutional and recurrent learning paradigms, enabling a comprehensive comparison of

spatial feature extraction and temporal sequence modeling capabilities for physiological waveform analysis.

Convolutional Neural Network (CNN)

Convolutional Neural Networks (**CNNs**) were used to capture local patterns and morphological features from physiological signals. CNNs are particularly effective in extracting meaningful information from waveform data due to their ability to automatically learn hierarchical feature representations. In this study, one-dimensional convolutional layers were employed to process ECG, PPG, and RESP signals directly from raw input sequences.

Recurrent Neural Network (RNN)

Recurrent Neural Networks (**RNNs**) were evaluated because they are designed to process sequential data and model temporal dependencies. RNNs maintain information from previous time steps, allowing them to capture relationships within physiological signal sequences.

Long Short-Term Memory (LSTM)

LSTM networks were included to overcome the vanishing gradient problem commonly encountered in traditional RNNs. LSTMs utilize memory cells and gating mechanisms that enable the model to learn long-term temporal dependencies, making them suitable for physiological time-series analysis.

Gated Recurrent Unit (GRU)

GRU networks provide a simplified alternative to LSTMs while maintaining the ability to model long-range temporal relationships. GRUs contain fewer parameters than LSTMs, resulting in lower computational complexity and faster training while preserving competitive performance.

Bidirectional Long Short-Term Memory (Bi-LSTM)

Bi-LSTM networks process input sequences in both forward and backward directions. This enables the model to utilize information from past and future time steps simultaneously, improving its ability to capture complex temporal patterns within physiological signals.

Multi-Task Learning

Blood pressure estimation was formulated as a multi-task regression problem. Instead of training separate models for systolic blood pressure (SBP) and diastolic blood pressure (DBP), a single model was trained to predict both values simultaneously. This approach allows the model to learn shared physiological representations and exploit the relationship between SBP and DBP.

Multimodal Fusion Models

To investigate the benefits of combining multiple physiological signals, several multimodal fusion strategies were evaluated, including Feature Concatenation, Bidirectional Cross Attention (**BCA**), Gated Multimodal Unit (**GMU**), and the proposed Dominant-Guided Auxiliary Gated Fusion (**DGAGF**) framework.

DGAGF was designed to treat PPG as the dominant modality while selectively incorporating complementary information from ECG and RESP signals through adaptive gating mechanisms. This approach reduces the impact of noisy or redundant modalities and improves multimodal blood pressure estimation performance.

The performance of all models was evaluated using identical preprocessing procedures, dataset splits, and training settings to ensure a fair and consistent comparison across modalities and architectures.

Model	Primary Purpose
CNN	Local feature extraction
RNN	Sequential pattern learning
LSTM	Long-term dependency modeling
GRU	Efficient temporal modeling
Bi-LSTM	Bidirectional sequence learning
DGAGF	normalization Multimodal feature fusion

3.5 Data Flow

The data flow within the SENTINEL system follows a structured pipeline that transforms raw physiological signals into blood pressure predictions and monitoring records.

The process begins when physiological signal data are obtained from the MIMIC-BP dataset or uploaded through the monitoring platform. The input signals, including ECG, PPG, and RESP, are first passed through the preprocessing stage, where synchronization, segmentation, and procedures are applied.

The processed signals are then forwarded to the deep learning prediction models. Depending on the experimental configuration, the system performs either single-modality or multimodal analysis. The trained models extract relevant physiological features and generate systolic blood pressure (SBP) and diastolic blood pressure (DBP) predictions.

The predicted values are subsequently evaluated and classified into blood pressure status categories, such as Normal, Low, or High. The generated results are stored in the database and made available through the monitoring platform.

Finally, users can access prediction results, health status information, and historical monitoring records through the web-based interface. This workflow enables seamless communication between the data source, preprocessing pipeline, prediction models, database, and user interface.

3.6 Dataset Description & Preprocessing

3.6.1 Dataset Source and Justification

The experiments were conducted using the MIMIC-BP dataset, a publicly available benchmark dataset designed specifically for cuffless blood pressure estimation [20]. The dataset was derived from the MIMIC waveform databases and contains synchronized physiological signals together with corresponding blood pressure labels [20].

MIMIC-BP was selected for this study because it is one of the few publicly available datasets that simultaneously provides ECG, PPG, RESP, and blood pressure measurements. In addition, the dataset was specifically curated to reduce data leakage and information shift, making it suitable for fair evaluation and comparison of machine learning and deep learning models [20].

3.6.2 Dataset Size and Format

The dataset contains physiological recordings from 1,524 subjects. The signals are organized into waveform segments associated with systolic blood pressure (SBP) and diastolic blood pressure (DBP) labels.

Each physiological segment has:

- Duration: 30 seconds
- Sampling Rate: 125 Hz
- Window Length: 3750 samples

The processed signals are stored as NumPy arrays and organized into fixed-length waveform segments that serve as input samples for model training and evaluation.

3.6.3 Dataset Features

Three physiological modalities were used in this study:

1. **Electrocardiogram (ECG)**
 1. Measures the electrical activity of the heart [5], [7].
 2. Provides information about cardiac rhythm and timing [5], [7].
2. **Photoplethysmogram (PPG)**
 1. Measures blood volume changes in peripheral blood vessels [3], [23].
 2. Strongly correlated with vascular dynamics and blood pressure variations [3], [23].
3. **Respiration (RESP)**
 1. Measures respiratory activity and breathing patterns [24]–[27].
 2. Provides complementary cardiovascular information through cardiorespiratory interactions [24]–[27].

Each sample is associated with two target variables:

- Systolic Blood Pressure (SBP)
- Diastolic Blood Pressure (DBP)

These labels serve as the ground-truth values for supervised learning [20].

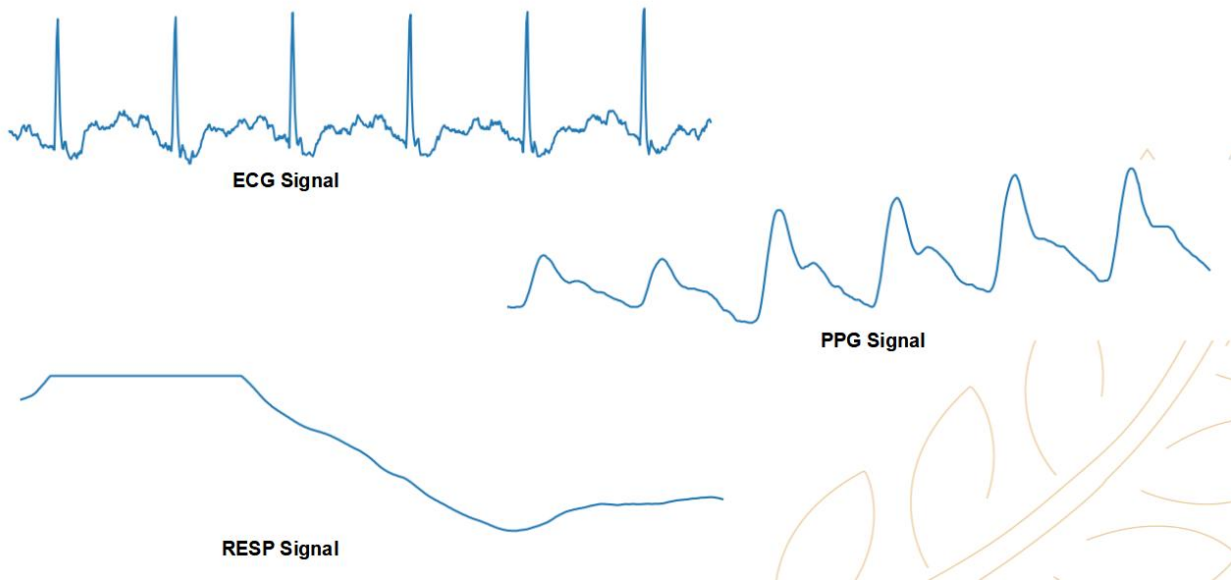


Figure 3.1: Example physiological signal samples from the MIMIC-BP dataset, including ECG, PPG, and RESP modalities.

These labels serve as the ground-truth values for supervised learning. The SBP and DBP values were used as continuous regression targets during model training and evaluation.

3.6.4 Data Cleaning and Labeling

The MIMIC-BP dataset provides curated physiological recordings and corresponding blood pressure labels [20]. Therefore, no additional signal filtering was applied beyond the preprocessing procedures adopted in this study. The physiological signals were synchronized with their corresponding blood pressure measurements and prepared for segmentation and normalization.

Each physiological segment was assigned its corresponding SBP and DBP values, which were used as labels during model training and evaluation.

3.6.5 Data Splitting

To prevent data leakage and ensure reliable model evaluation, a subject-level splitting strategy was adopted. The dataset was divided into:

- Training Set: 70%
- Validation Set: 10%
- Testing Set: 20%

Subject-level separation guarantees that records belonging to the same subject do not appear in multiple subsets.

3.6.6 Preprocessing Steps

A unified preprocessing pipeline was applied to all physiological modalities to ensure fair comparison across different models and signal types.

The preprocessing process consisted of the following steps:

1. Signal Synchronization

- ECG, PPG, RESP, and blood pressure labels were temporally aligned.

2. Segmentation

- Continuous signals were divided into fixed-length windows of 30 seconds.

3. Sample Preparation

- Each segment was treated as an independent training sample.

4. Z-Score Normalization

- Signal amplitudes were normalized using Z-score normalization to reduce variability and improve training stability.

5. Dataset Preparation

- Processed samples were organized into training, validation, and testing sets for model development and evaluation.

The overall preprocessing pipeline applied in this study is illustrated in Figure 3.



Figure 3.2: Preprocessing pipeline applied to ECG, PPG, and RESP signals before model training and evaluation.

This standardized preprocessing pipeline ensures that performance differences are primarily attributable to modality selection and model architecture rather than differences in data preparation.

3.7 Design Rationale

Several design decisions were made throughout the development of the proposed SENTINEL system to ensure accurate blood pressure estimation, fair model comparison, and practical deployment for real-world monitoring applications.

First, the MIMIC-BP dataset was selected because it provides synchronized ECG, PPG, and RESP signals together with corresponding blood pressure labels [20]. Unlike many publicly available datasets, MIMIC-BP supports both single-modality and multimodal blood pressure estimation while reducing the risk of subject-level data leakage [20].

In addition, the availability of multiple physiological modalities within the same dataset enabled comprehensive investigation of both single-modality and multimodal learning approaches under consistent experimental conditions.

A multi-task learning approach was adopted instead of training separate models for systolic blood pressure (SBP) and diastolic blood pressure (DBP). Since both targets are physiologically related, joint learning allows the model to learn shared representations and improve prediction efficiency while reducing computational complexity.

Multiple deep learning architectures, including CNN, RNN, LSTM, GRU, and Bi-LSTM, were evaluated rather than relying on a single model. This decision enabled a fair comparison of different learning approaches and helped identify the most suitable architecture for each physiological modality.

For multimodal learning, several fusion strategies were investigated. Traditional fusion methods such as feature concatenation and attention-based fusion were considered; however, these approaches may treat all modalities equally and may not effectively handle noisy or redundant information [6], [12]. Therefore, the proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework was adopted to allow the dominant modality to guide the contribution of auxiliary modalities through adaptive gating mechanisms.

A unified preprocessing pipeline was applied to all modalities to ensure that performance differences resulted from model architecture and modality selection rather than preprocessing variations. Subject-level data splitting was also adopted to prevent information leakage between training and testing sets and provide a more reliable evaluation of model performance.

Finally, a web-based and mobile-responsive monitoring platform was selected to improve accessibility and usability. Compared with standalone machine learning models, integrating the prediction system into a complete monitoring platform enables practical deployment, continuous monitoring, historical record tracking, and easier interaction for patients and relatives.

Collectively, these design decisions contributed to the development of a reliable, scalable, and experimentally rigorous framework for cuffless blood pressure estimation and monitoring.

3.8 Implementation Plan

The implementation of the proposed SENTINEL system was carried out through multiple development phases to ensure a structured transition from system design to a fully functional blood pressure monitoring platform. Each phase focused on a specific component of the project while maintaining integration with the overall system architecture.

Phase 1: Dataset Preparation and Analysis

The first phase focused on understanding and preparing the MIMIC-BP dataset. The physiological signals and blood pressure labels were analyzed to identify the available

modalities, data structure, and target variables. Dataset statistics and signal characteristics were examined to support subsequent model development.

Phase 2: Data Preprocessing

In the second phase, a unified preprocessing pipeline was implemented. Signal synchronization, segmentation into fixed-length windows, and Z-score normalization were applied to prepare the data for training. Subject-level data splitting was also performed to create training, validation, and testing datasets while preventing data leakage.

Phase 3: Single-Modality Model Development

The third phase involved developing and evaluating multiple deep learning architectures, including CNN, RNN, LSTM, GRU, and Bi-LSTM. Each modality (ECG, PPG, and RESP) was evaluated independently to identify the most effective architecture for blood pressure estimation.

Phase 4: Multimodal Fusion Development

After identifying the best-performing single-modality models, multimodal learning experiments were conducted. Different fusion strategies, including Concatenation, Bidirectional Cross Attention (BCA), Gated Multimodal Unit (GMU), and the proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework, were implemented and evaluated.

Phase 5: Model Evaluation and Optimization

The developed models were trained and evaluated using standardized experimental settings. Performance was assessed using Mean Absolute Error (MAE), Mean Error (ME), and Standard Deviation (SD). Hyperparameter tuning and model selection were performed to identify the best-performing configurations.

Phase 6: Backend Development

A backend service was developed using FastAPI to provide communication between users and the prediction models. The backend handles signal processing, model inference, prediction generation, and data management.

Phase 7: Database Integration

A database layer was integrated to store prediction results, patient records, and monitoring history. This component enables long-term tracking and retrieval of historical blood pressure information.

Phase 8: Web and Mobile Interface Development

A responsive user interface was developed to support both desktop and mobile devices. The platform provides patient and relative portals, blood pressure prediction services, historical monitoring, and health status visualization.

Phase 9: System Testing and Deployment

The final phase focused on integrating all system components, validating functionality, testing prediction workflows, and ensuring stable operation of the complete SENTINEL monitoring platform.

These development phases ensured systematic implementation of the proposed framework, from data preparation and model development to deployment within the SENTINEL monitoring platform.

3.9 Software Requirements (Mandatory CS Subsection)

ID	Functional Requirement	Priority
FR-01	Patient can register an account	High
FR-02	Patient can log in to the system	High
FR-03	Relative can register an account	High
FR-04	Relative can log in to the system	High
FR-05	Relative can link to a patient using Patient ID	High
FR-06	User can upload ECG, PPG, or RESP signal files	High
FR-07	System preprocesses uploaded signals automatically	High
FR-08	System predicts SBP and DBP using AI models	High
FR-09	System displays blood pressure status (Normal, Low, High)	High
FR-10	System stores and displays prediction history	High

Non-Functional Requirements & Dependencies:

Usability

The system shall provide a simple and user-friendly interface that can be easily used by elderly patients and their relatives.

Performance

The system shall generate blood pressure predictions within a reasonable response time after signal upload.

Reliability

The system shall provide stable prediction services and maintain access to historical monitoring records.

Accuracy

The system shall provide reliable blood pressure estimation results through trained and validated deep learning models evaluated using standard regression metrics such as MAE, ME, and SD.

Maintainability

The system shall be modularly designed to facilitate future updates, model replacement, and feature extension.

Scalability

The system shall support the integration of additional physiological modalities, prediction models, and future monitoring features.

Security

User authentication shall be required before accessing personal monitoring information.

System Constraints

- The system currently depends on physiological signals available in the MIMIC-BP dataset.
- Prediction performance depends on the quality of uploaded physiological signals.
- The current implementation focuses on blood pressure estimation and monitoring only.
- Internet access is required to use the web platform.

Assumptions

- Uploaded signal files are correctly formatted and compatible with the system requirements.
- Users possess valid patient credentials before accessing monitoring services.
- Physiological signals contain sufficient information for blood pressure estimation.

Software Tools and Libraries

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- FastAPI
- Python

Machine Learning

- TensorFlow
- Keras
- NumPy
- Pandas
- Scikit-learn

Database

- SQLite

Development Tools

- Visual Studio Code
- Google Colab

- Kaggle

Hardware Dependencies

Training Environment

- NVIDIA Tesla T4 GPU
- Intel Xeon CPU
- 12 GB RAM

Deployment Environment

- Standard desktop or laptop computer
- Modern web browser
- Internet connection

3.10 Minimal Project Management (Mandatory CS Subsection)

Milestone / Task	Owner	Timeline	Status
Requirements Analysis & Literature Review	Joint	Weeks 1–5	Completed
System Design & Software Architecture	Rawan	Weeks 6–9	Completed
Dataset Preparation & Signal Preprocessing	Joint	Weeks 8–11	Completed
Single-Modality AI Model Development	Menna	Weeks 10–13	Completed
Multimodal Fusion Model Development	Joint	Weeks 10–18	Completed
Frontend Development	Rawan	Weeks 20–25	Completed
Backend Development	Rawan	Weeks 20–26	Completed
Database Design & Integration	Rawan	Weeks 22–26	Completed
AI Model Integration	Joint	Weeks 30–33	Completed
System Testing & Validation	Joint	Weeks 33–36	Completed
Documentation & Thesis Preparation	Menna	Weeks 34–38	Completed

Risks & Mitigation:

Risk	Likelihood / Impact	Mitigation
Data Leakage between training and testing sets	High	Subject-level data splitting was applied

Risk	Likelihood / Impact	Mitigation
Noisy physiological signals	Medium	Signal preprocessing and normalization were performed
Model overfitting	Medium	Validation monitoring and regularization techniques were used
Integration issues between AI model and web platform	Medium	Modular system design and continuous testing were adopted

Chapter 4: Implementation

4.1 Tools & Technologies

The SENTINEL system was developed using a combination of machine learning, web development, database management, and deployment technologies. The selected tools and frameworks were chosen to support physiological signal processing, deep learning model development, backend services, and user interface implementation.

The technology stack was selected to support the complete development lifecycle of the project, including data preprocessing, model training, prediction serving, database management, and user interaction through a web-based monitoring platform.

Programming Languages

- Python
- JavaScript
- HTML5
- CSS3

Python was used for data preprocessing, model development, training, evaluation, and backend implementation. JavaScript, HTML, and CSS were used for developing the web-based user interface.

Machine Learning Frameworks and Libraries

- TensorFlow
- Keras
- NumPy
- Pandas
- Scikit-learn

TensorFlow and Keras were used to implement and train the deep learning models. NumPy and Pandas were utilized for numerical computation and data manipulation, while Scikit-learn supported data preprocessing and evaluation tasks.

Backend Technologies

- FastAPI
- Uvicorn

FastAPI was used to develop the backend API responsible for receiving uploaded physiological signals, performing model inference, and returning prediction results. Uvicorn was used as the ASGI server for running the FastAPI application.

Frontend Technologies

- React.js
- HTML5
- CSS3
- JavaScript

React.js was used to develop a responsive and interactive user interface for patients and relatives. The frontend communicates with the backend API to upload signals and display prediction results.

Database Technologies

- SQLite

SQLite was used to store prediction results, patient records, and monitoring history due to its lightweight design and ease of integration.

Development Platforms and Tools

- Visual Studio Code
- Google Colab
- Kaggle

Visual Studio Code was used as the primary development environment. Google Colab and Kaggle were utilized for model training and experimentation using GPU resources.

4.2 Development Environment

The development and experimentation processes of the SENTINEL system were conducted using a combination of local and cloud-based environments to support software development and deep learning model training.

Hardware Environment

The deep learning experiments were performed using cloud-based GPU resources. NVIDIA Tesla T4 GPUs, Intel Xeon processors, and approximately 12 GB RAM were utilized to accelerate model training and evaluation.

Software Environment

The project was developed using Windows 10/11 as the primary operating system. Python was used for data preprocessing, model development, model evaluation, and backend implementation. JavaScript was used for frontend development and user interface functionality.

Integrated Development Environment (IDE)

Visual Studio Code (VS Code) was used as the primary integrated development environment for implementing and managing the project components. The environment provided support for Python, JavaScript, debugging tools, and project organization.

Development and Training Platforms

Google Colab and Kaggle were used for training and evaluating the deep learning models. These platforms provided access to GPU resources and facilitated the execution of computationally intensive experiments.

Deployment Setup

The implemented system follows a three-tier architecture consisting of a React-based frontend, a FastAPI backend service, and an SQLite database. The frontend provides user interaction and visualization capabilities, while the backend handles signal processing, model inference, and communication with the database. The database stores prediction records and monitoring history for future retrieval and analysis.

4.3 System Modules

The SENTINEL platform was implemented as a modular system consisting of several interconnected components. Each module is responsible for a specific functionality within the blood pressure monitoring workflow, ensuring maintainability, scalability, and efficient system operation.

Authentication Module

The Authentication Module manages user registration and login operations. The system supports both patient and relative accounts, allowing authorized users to securely access the monitoring platform. User credentials are verified before granting access to the corresponding dashboard and system services.

Signal Upload Module

The Signal Upload Module allows users to upload physiological signal files for blood pressure estimation. The module accepts ECG, PPG, and RESP signal files depending on the selected configuration. Uploaded files are validated before being forwarded to the prediction pipeline.

Signal Processing Module

The Signal Processing Module prepares uploaded physiological signals for model inference. This module performs signal formatting, validation, reshaping, and normalization to ensure compatibility with the trained deep learning models.

Blood Pressure Prediction Module

The Blood Pressure Prediction Module serves as the core intelligence component of the system. The module loads the trained deep learning models and generates systolic blood pressure (SBP) and diastolic blood pressure (DBP) predictions from the uploaded physiological signals. Based on the predicted values, the system determines the corresponding blood pressure status.

Monitoring and History Module

The Monitoring and History Module stores prediction results and allows users to access previously generated records. Historical monitoring information enables patients and relatives to review blood pressure trends and past prediction outcomes.

Database Module

The Database Module is responsible for storing prediction records, patient identifiers, blood pressure values, status classifications, and timestamps. SQLite was used to provide lightweight and efficient data management.

User Interface Module

The User Interface Module provides an interactive web-based environment through which users interact with the system. The interface supports signal upload, prediction visualization, status display, and monitoring history retrieval through responsive dashboard pages accessible from desktop and mobile devices.

The modular architecture facilitates independent development, testing, and maintenance of system components while ensuring seamless interaction between prediction services, data management, and user-facing functionalities.

4.4 Algorithms / Models Implemented

Several deep learning models were implemented and evaluated to estimate systolic blood pressure (SBP) and diastolic blood pressure (DBP) from physiological signals. The implemented models were designed to process ECG, PPG, and RESP waveforms and learn the relationship between physiological patterns and blood pressure values.

Convolutional Neural Network (CNN)

The CNN model was implemented using one-dimensional convolutional layers to process physiological time-series signals. Multiple convolutional blocks were employed to automatically extract hierarchical features from the input waveforms. Batch normalization, activation functions, pooling layers, and dropout regularization were incorporated to improve training stability and reduce overfitting. The extracted features were forwarded to fully connected layers that generated SBP and DBP predictions.

Long Short-Term Memory (LSTM)

The LSTM model was implemented to capture temporal dependencies within physiological signals. LSTM memory cells and gating mechanisms enabled the model to learn long-term relationships between waveform patterns and blood pressure values. The extracted temporal features were connected to a regression head for simultaneous SBP and DBP prediction.

Recurrent Neural Network (RNN)

The RNN architecture was implemented to model sequential relationships within physiological waveform data. The recurrent structure allowed information from previous time steps to influence future predictions, enabling the model to learn temporal signal characteristics.

Gated Recurrent Unit (GRU)

The GRU model was implemented as a computationally efficient alternative to LSTM. The simplified gating mechanism reduced the number of trainable parameters while maintaining the ability to capture temporal dependencies within physiological signals.

Bidirectional Long Short-Term Memory (Bi-LSTM)

The Bi-LSTM model was implemented to process physiological sequences in both forward and backward directions. This architecture enabled the model to utilize information from past and future time steps simultaneously, improving temporal feature representation.

Multi-Task Learning Head

All implemented models employed a multi-task learning framework to jointly predict systolic blood pressure (SBP) and diastolic blood pressure (DBP). Instead of training separate models for each target, a shared feature extraction backbone was connected to a regression head that simultaneously estimated both blood pressure values.

Dominant-Guided Auxiliary Gated Fusion (DGAGF)

For multimodal learning, the proposed DGAGF framework was implemented to combine information from ECG, PPG, and RESP signals. The framework treats PPG as the dominant modality while ECG and RESP serve as auxiliary modalities. Adaptive gating mechanisms were used to control the contribution of each modality during feature fusion, enabling the model to emphasize informative features while reducing the influence of redundant information.

Training Configuration

All implemented models were trained using the Adam optimizer and Huber loss function. Model performance was evaluated using Mean Absolute Error (MAE), Mean Error (ME), and Standard Deviation (SD). A subject-level train-validation-test split was applied to ensure fair evaluation and prevent data leakage.

Parameter	Value
Optimizer	Adam
Loss Function	Huber Loss
Sampling Rate	125 Hz
Window Length	3750
Input Duration	30 s
Output Targets	SBP, DBP

4.5 Database / API Details

The SENTINEL platform utilizes a database layer and RESTful API services to manage communication between the user interface, prediction models, and stored monitoring records.

The integration layer serves as the communication bridge between the web interface, the trained deep learning models, and the database, enabling automated prediction and monitoring workflows.

Database Schema

SQLite was used as the database management system due to its lightweight architecture and ease of integration. The database stores prediction records generated by the blood pressure estimation models and maintains monitoring history for future retrieval.

The primary prediction table contains the following fields:

Field	Data Type	Description
patient_id	TEXT	Unique patient identifier
sbp	REAL	Predicted systolic blood pressure
dbp	REAL	Predicted diastolic blood pressure
status	TEXT	Blood pressure status (Normal, Low, High)
timestamp	DATETIME	Prediction date and time

This structure enables efficient storage and retrieval of monitoring records for both patients and relatives.

API Integration

FastAPI was used to implement RESTful API services that connect the frontend interface with the prediction models and database layer.

The implemented API endpoints include:

Endpoint	Method	Description
/	GET	Returns system status and model availability information
/predict	POST	Receives physiological signal files and generates blood pressure predictions
/history/{patient_id}	GET	Retrieves monitoring history for a specific patient
/patients	GET	Returns available patient records and monitoring statistics

Prediction Workflow

When a user uploads a physiological signal file through the web interface, the file is transmitted to the backend API. The backend validates and preprocesses the signal before forwarding it to the trained deep learning model. The model generates systolic blood pressure (SBP) and diastolic blood pressure (DBP) predictions, which are subsequently classified into blood pressure status categories. The generated results are stored in the database and returned to the frontend for visualization.

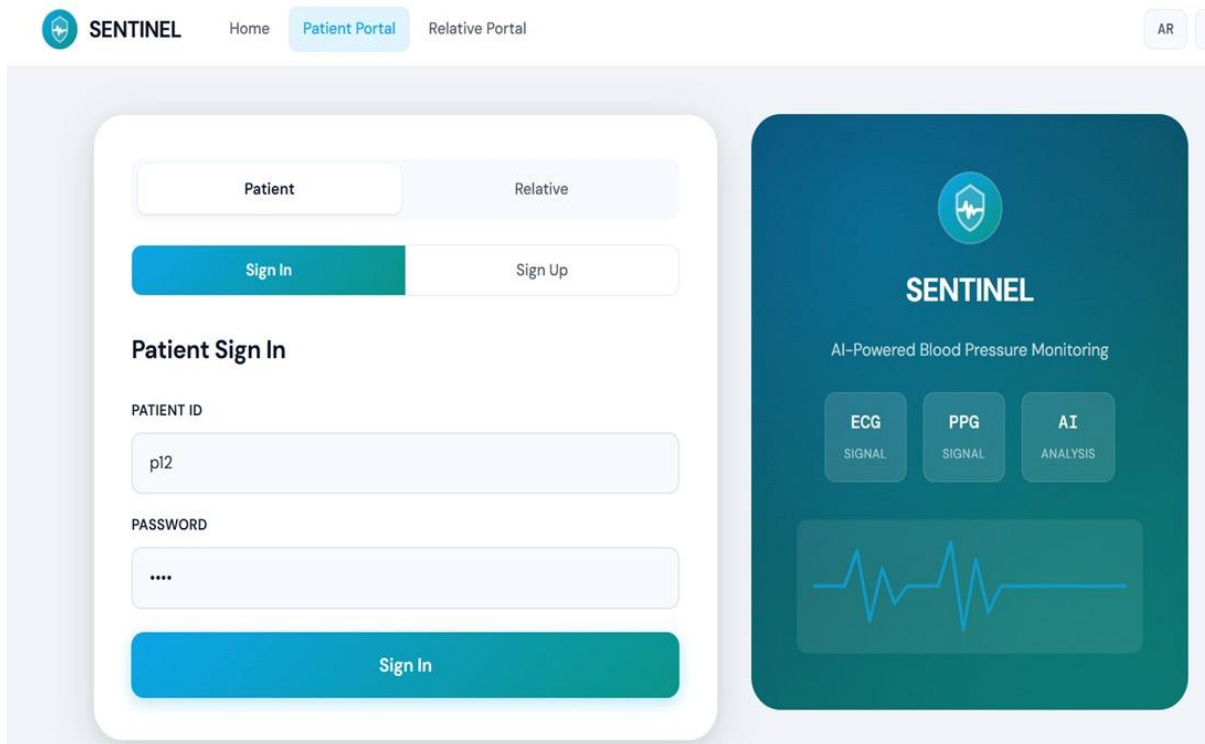
This integration enables seamless communication between the user interface, machine learning models, and database services, supporting real-time blood pressure estimation and monitoring.

4.6 Screenshots & Diagrams

This section presents screenshots of the implemented SENTINEL platform. The figures illustrate the main interfaces developed for patients and relatives, including signal upload, blood pressure prediction, monitoring history, and API testing.

Figure 4.1 Patient Sign In Interface

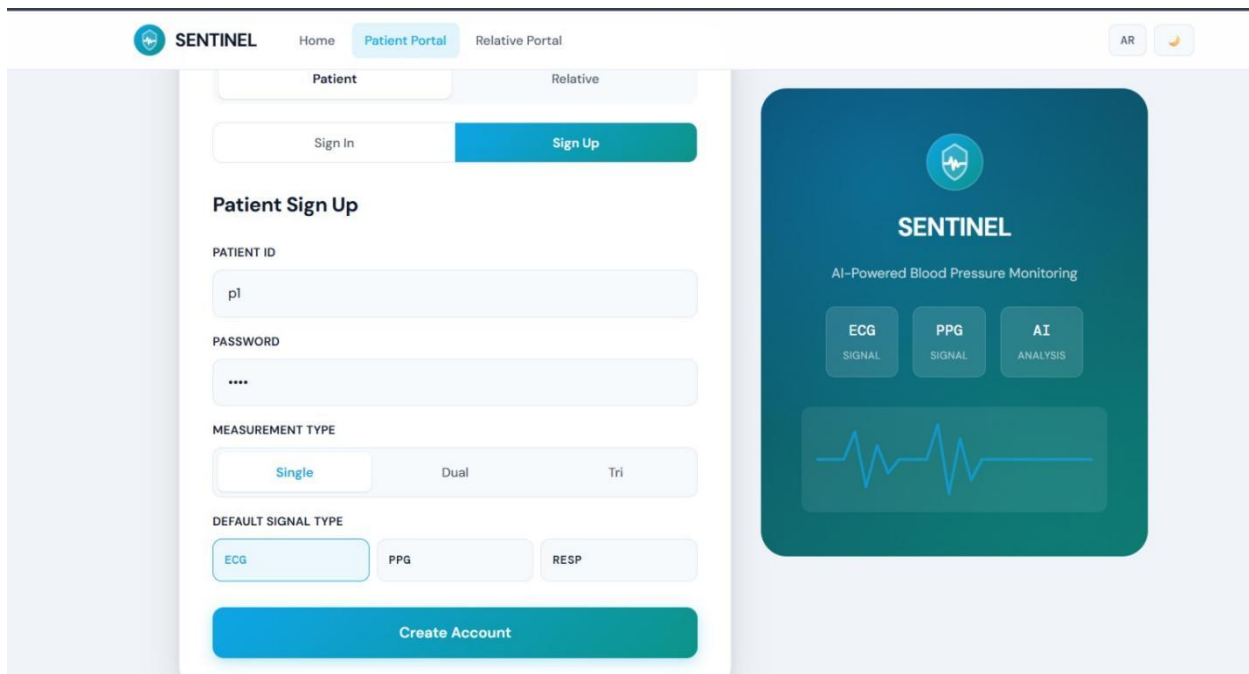
Patient login interface used to authenticate patients before accessing blood pressure monitoring services.



The image shows the 'Patient Sign In' interface of the SENTINEL application. The top navigation bar includes the SENTINEL logo, 'Home', 'Patient Portal' (active), and 'Relative Portal'. On the right, there are 'AR' and a user profile icon. The main content area is divided into two panels. The left panel has tabs for 'Patient' and 'Relative'. Under the 'Patient' tab, there are 'Sign In' and 'Sign Up' buttons. The 'Sign In' button is highlighted. Below the buttons, the 'Patient Sign In' section contains a 'PATIENT ID' field with the value 'p12' and a 'PASSWORD' field with four dots. A large blue 'Sign In' button is at the bottom. The right panel features the SENTINEL logo, the text 'AI-Powered Blood Pressure Monitoring', and three buttons: 'ECG SIGNAL', 'PPG SIGNAL', and 'AI ANALYSIS'. A blue ECG waveform is displayed at the bottom of this panel.

Figure 4.2 Patient Registration Interface

Patient registration interface allowing users to create accounts and select the preferred signal configuration.



The image shows the 'Patient Sign Up' interface of the SENTINEL application. The top navigation bar is identical to Figure 4.2. The main content area is divided into two panels. The left panel has tabs for 'Patient' and 'Relative'. Under the 'Patient' tab, there are 'Sign In' and 'Sign Up' buttons. The 'Sign Up' button is highlighted. Below the buttons, the 'Patient Sign Up' section contains a 'PATIENT ID' field with the value 'p1', a 'PASSWORD' field with four dots, a 'MEASUREMENT TYPE' section with 'Single', 'Dual', and 'Tri' options (where 'Single' is selected), and a 'DEFAULT SIGNAL TYPE' section with 'ECG', 'PPG', and 'RESP' options (where 'ECG' is selected). A large blue 'Create Account' button is at the bottom. The right panel is identical to the one in Figure 4.2, featuring the SENTINEL logo, 'AI-Powered Blood Pressure Monitoring', three buttons for 'ECG SIGNAL', 'PPG SIGNAL', and 'AI ANALYSIS', and an ECG waveform.

Figure 4.3 Relative Registration Interface

Relative registration interface used to link relatives with patients through the patient identifier

 SENTINEL

HomePatient PortalRelative Portal

AR

PatientRelative

Sign InSign Up

Relative Sign Up

LINKED PATIENT ID

p1

RELATIVE NAME

Rawan

RELATIONSHIP

Daughter

PHONE NUMBER

01229015829

PASSWORD

....

Create Relative Account

 SENTINEL

AI-Powered Blood Pressure Monitoring

ECG SIGNALPPG SIGNALAI ANALYSIS



Figure 4.4 Patient Signal Upload Interface

Signal upload interface allowing patients to upload physiological signal files for blood pressure prediction.

 SENTINEL

Patient Portal

AR Sign Out

 WELCOME BACK

Patient p12

REQUIRED SIGNAL PPG

Upload Signal



✓ p000801_ppg.npy

 Check My Blood Pressure

Figure 4.5 Relative Monitoring Dashboard

Relative monitoring dashboard displaying patient alerts, blood pressure statistics, and monitoring information.

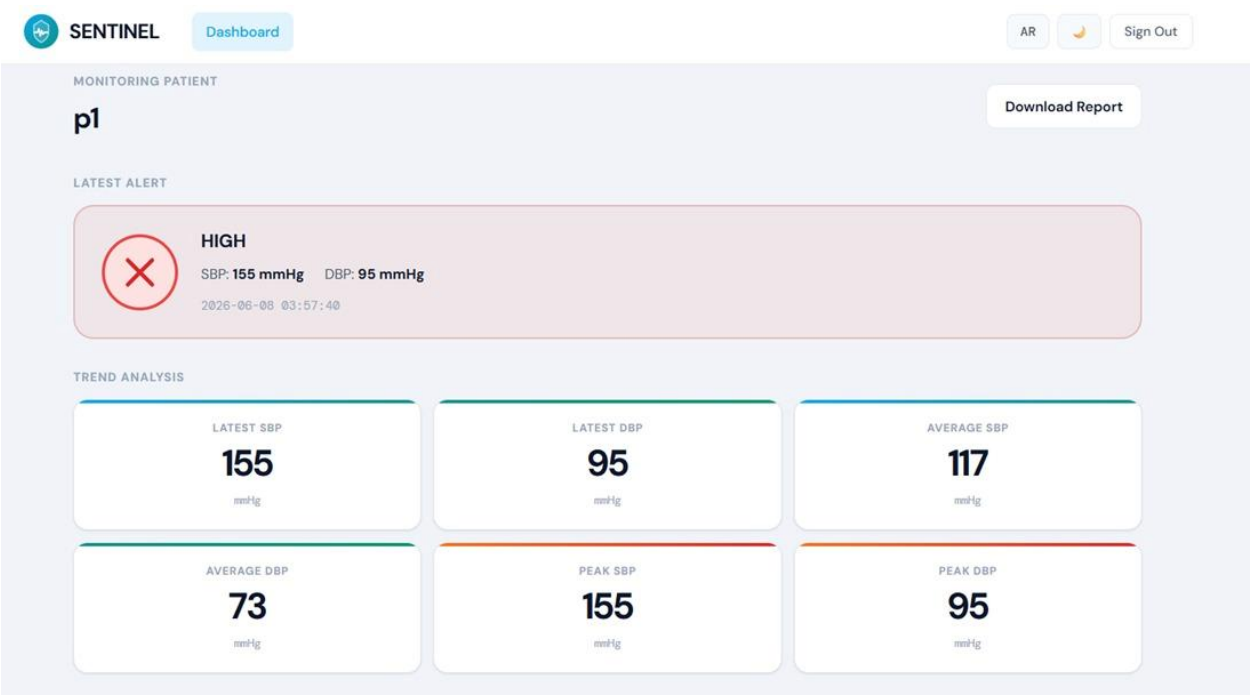


Figure 4.6 Blood Pressure Trend Dashboard

Blood pressure trend visualization showing historical SBP and DBP monitoring records.



Figure 4.7 API Testing Results

API testing results demonstrating successful execution of backend endpoints and system functionality.

All Tests	Passed (24)	Failed (0)	Skipped (0)	Errors (0)	Console Log	View Summary
PASS	relative login successful					
GET	04 Monitoring & History / TC-06 Patient History Positive					
http://127.0.0.1:8000/history/p100	200	4 ms	442 B	2		
PASS	Status code is 200					
PASS	History returned successfully					
GET	04 Monitoring & History / TC-07 Get Patients Positive					
http://127.0.0.1:8000/patients	200	3 ms	217 B	2		
PASS	Status code is 200					
PASS	Patients list returned successfully					
GET	04 Monitoring & History / TC-08 Relative Latest Alert Positive					
http://127.0.0.1:8000/relative-alert/p100/Ahmed	200	5 ms	244 B	2		
PASS	Status code is 200					
PASS	Relative alert returned successfully					
GET	04 Monitoring & History / TC-09 Relative History Positive					
http://127.0.0.1:8000/relative-history/p100/Ahmed	200	5 ms	466 B	2		
PASS	Status code is 200					
PASS	Relative history returned successfully					

Chapter 5: Results, Testing, and Evaluation

5.1 Experimental Setup

This section describes the experimental setup used to evaluate the proposed blood pressure estimation framework. All experiments were conducted under identical training and evaluation conditions to ensure fair comparison between different physiological modalities, model architectures, and fusion strategies.

Experimental Environment

Model development and training were performed using cloud-based GPU resources. The experiments utilized NVIDIA Tesla T4 GPUs with Intel Xeon processors and approximately 12 GB RAM. Training and evaluation were conducted using Google Colab and Kaggle environments.

Training Configuration

Five deep learning architectures were evaluated, including Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and Bidirectional Long Short-Term Memory (Bi-LSTM). All models were implemented within a multi-task learning framework to jointly predict systolic blood pressure (SBP) and diastolic blood pressure (DBP).

A unified preprocessing pipeline was applied to all experiments. Physiological signals were synchronized, segmented into fixed-length windows, normalized using Z-score normalization, and divided using a subject-level train-validation-test split to prevent data leakage. The experiments were conducted using the MIMIC-BP dataset [20].

Evaluation Protocol

Model performance was assessed using Mean Absolute Error (MAE), Mean Error (ME), and Standard Deviation (SD). These metrics were computed for both SBP and DBP predictions and were used to compare different model architectures and physiological modalities.

Experimental Phases

The evaluation process consisted of two main phases. The first phase focused on single-modality learning using ECG, PPG, and RESP signals independently. The second phase investigated multimodal learning using different modality combinations, including ECG+PPG, ECG+RESP, PPG+RESP, and ECG+PPG+RESP.

For multimodal experiments, several fusion strategies were evaluated, including Feature Concatenation, Bidirectional Cross Attention (BCA), Gated Multimodal Unit (GMU), and the proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework. The selected fusion methods were motivated by previous multimodal blood pressure estimation studies [6], [9], [12].

5.2 Results

This section presents the experimental results obtained from the proposed blood pressure estimation framework. The evaluation was conducted using both single-modality and multimodal physiological signals to assess the effectiveness of different deep learning architectures and fusion strategies. Model performance was measured using standard regression metrics, including Mean Absolute Error (MAE),

Mean Error (ME), and Standard Deviation (SD). The results are presented through tables and figures to facilitate comparison between modalities, architectures, and fusion approaches.

5.2.1 Single-Modality Results

Table 5.1: Single-Modality Blood Pressure Estimation Results Using ECG, PPG, and RESP Signals.

Modality	Model	SBP			DBP			Overall		
		Best	Avg.	STD	Best	Avg.	STD	Best	Avg.	STD
ECG	CNN	11.90	13.29	1.274	9.33	9.28	0.062	21.23	22.57	1.230
	LSTM	11.78	13.20	1.277	9.40	9.38	0.040	21.18	22.58	1.255
	RNN	11.79	13.24	1.296	9.42	9.41	0.023	21.21	22.65	1.275
	GRU	11.76	13.20	1.278	9.40	9.40	0.024	21.16	22.60	1.254
	Bi-LSTM	11.69	13.18	1.358	9.26	9.34	0.080	20.95	22.52	1.435
PPG	CNN	<u>10.45</u>	<u>11.74</u>	<u>1.174</u>	<u>8.23</u>	<u>8.22</u>	0.022	<u>18.68</u>	<u>19.95</u>	<u>1.162</u>
	LSTM	11.72	13.17	1.311	9.30	9.33	0.011	21.02	22.49	1.300
	RNN	11.73	13.16	1.309	9.33	9.32	0.009	21.06	22.49	1.304
	GRU	11.74	13.18	1.311	9.33	9.33	<u>0.008</u>	21.07	22.50	1.309
	Bi-LSTM	11.66	13.12	1.337	9.25	9.28	0.079	20.91	22.40	1.369
RESP	CNN	11.80	13.22	1.293	9.44	9.45	<u>0.008</u>	21.24	22.66	1.291
	LSTM	11.78	13.23	1.320	9.37	9.38	0.023	21.15	22.60	1.328
	RNN	11.78	13.22	1.316	9.36	9.40	0.038	21.14	22.62	1.351
	GRU	11.78	13.22	1.318	9.41	9.42	0.009	21.19	22.64	1.324
	Bi-LSTM	11.69	13.19	1.368	9.26	9.36	0.092	20.95	22.55	1.459

The results of the single-modality experiments are presented in Table 5.1. The evaluation was conducted using ECG, PPG, and RESP signals across five deep learning architectures: CNN, LSTM, RNN, GRU, and Bi-LSTM.

The results reveal noticeable performance differences among the evaluated physiological modalities and model architectures. Among all modalities, PPG consistently achieved the best overall performance. The CNN model using PPG signals produced the lowest overall average error of 19.95, representing the best result among all single-modality configurations.

Although ECG and RESP also demonstrated competitive performance, their overall prediction errors were generally higher than those obtained using PPG. This suggests that PPG provides more direct and informative representations of cardiovascular dynamics that are closely related to blood pressure variations.

Overall, the findings indicate that PPG is the most informative individual modality for cuffless blood pressure estimation in this study, while CNN-based architectures are particularly effective in extracting blood-pressure-related features from physiological waveform signals.

5.2.2 Multimodal Results

Table 5.2: Multimodal Blood Pressure Estimation Results Using Different Fusion Strategies.

Data	Model	SBP			DBP			Overall	
		Best	Avg.	STD	Best	Avg.	STD	Best	Avg.
PPG+ECG	Concat.	14.19	14.26	0.072	9.28	9.27	0.083	23.47	23.53
	BCA	14.16	14.18	0.015	9.25	9.29	0.028	23.41	23.47
	GMU	14.12	14.20	0.099	9.23	9.28	0.098	23.36	23.48
	Ours	14.15	14.20	0.188	9.21	9.33	0.090	23.36	23.53
PPG+RESP	Concat.	14.11	14.17	0.049	9.39	9.36	0.041	23.50	23.54
	BCA	14.17	14.38	0.172	9.40	9.40	0.046	23.57	23.78
	GMU	12.73	12.87	0.180	8.09	8.18	0.116	20.83	21.05
	Ours	12.59	12.76	0.102	8.17	8.23	0.184	20.77	20.99
All	Concat.	14.09	14.27	0.125	9.28	9.32	0.078	23.37	23.60
	BCA	13.85	13.90	0.038	8.95	8.99	0.038	22.80	22.89
	GMU	15.37	16.11	0.681	9.78	9.82	0.163	25.15	25.93
	Ours	12.73	12.79	0.111	8.13	8.26	0.088	20.86	21.05

The multimodal experimental results are summarized in Table 5.2. Different modality combinations and fusion strategies were evaluated, including Feature Concatenation, Bidirectional Cross Attention (BCA), Gated Multimodal Unit (GMU), and the proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework.

The results demonstrate that combining multiple physiological signals generally improved blood pressure estimation performance compared with single-modality learning. The multimodal approaches benefited from the complementary information provided by different physiological modalities, leading to lower prediction errors and more stable performance.

Among the evaluated fusion strategies, the proposed DGAGF framework achieved competitive or superior performance across the investigated modality combinations. For the PPG+RESP configuration, DGAGF achieved the lowest overall average error of 20.99, outperforming the traditional fusion approaches. Similarly, in the three-modality configuration (ECG + PPG + RESP), DGAGF achieved the best overall average error of 21.05, demonstrating its effectiveness in integrating complementary physiological information for cuffless blood pressure estimation.

These findings indicate that adaptive fusion mechanisms can effectively exploit the strengths of multiple physiological signals and contribute to improved blood pressure prediction performance compared with conventional fusion methods.

5.3 Evaluation Metrics & Analysis

The performance of the proposed blood pressure estimation framework was evaluated using three standard regression metrics: Mean Absolute Error (MAE), Mean Error (ME), and Standard Deviation (SD). These metrics were selected because they provide a comprehensive assessment of prediction accuracy, bias, and consistency.

Mean Absolute Error (MAE) measures the average magnitude of prediction errors regardless of their direction. Lower MAE values indicate more accurate blood pressure estimation. Mean Error (ME) evaluates whether the model systematically overestimates or underestimates blood pressure values.

Standard Deviation (SD) measures the variability of prediction errors and reflects the stability and reliability of the model.

The experimental results demonstrated that physiological modality selection has a significant impact on blood pressure estimation performance. Among the evaluated single-modality configurations, PPG consistently outperformed ECG and RESP signals. This can be attributed to the strong physiological relationship between PPG waveforms and blood volume changes, which are directly influenced by blood pressure dynamics.

The results also revealed that CNN-based architectures achieved the best overall performance among the evaluated deep learning models. CNNs effectively capture local temporal patterns and morphological characteristics within physiological waveforms, enabling more accurate extraction of blood-pressure-related features.

Furthermore, multimodal learning improved estimation performance compared with single-modality approaches. By combining information from multiple physiological signals, the models were able to exploit complementary cardiovascular and respiratory characteristics that are not fully represented by a single modality.

Among the investigated fusion strategies, the proposed Dominant-Guided Auxiliary Gated Fusion (DGAGF) framework achieved the most competitive performance. The adaptive gating mechanism allowed the dominant modality to guide the contribution of auxiliary modalities, reducing the influence of redundant or less informative features while preserving complementary physiological information.

Overall, the experimental findings confirm that both modality selection and fusion strategy play a critical role in cuffless blood pressure estimation. The results demonstrate that combining informative physiological signals with adaptive multimodal fusion techniques can significantly enhance prediction accuracy and robustness.

5.4 Comparison with Baseline / Related Work

The findings obtained in this study were compared with several recent works on cuffless blood pressure estimation. Previous studies such as [10] demonstrated that PPG-based deep learning models can achieve promising blood pressure estimation performance using convolutional neural networks. Similarly, the results of the present study showed that PPG was the most informative physiological modality among the evaluated single-modality configurations, achieving the lowest overall prediction error.

Studies such as [9] investigated the combined use of ECG and PPG signals for blood pressure estimation and reported improved performance compared with single-signal approaches. The results obtained in this work support this observation, as multimodal learning generally outperformed single-modality models by exploiting complementary information from multiple physiological signals.

Recent multimodal approaches such as [12] highlighted the benefits of integrating ECG, PPG, and RESP signals for cuffless blood pressure estimation. Consistent with these findings, the proposed DGAGF framework demonstrated competitive multimodal performance by adaptively combining information from multiple physiological modalities.

Transformer-based approaches such as [11] demonstrated the ability of attention mechanisms to capture complex temporal relationships within physiological signals. However, such approaches often introduce higher computational complexity. In contrast, the proposed framework focuses on balancing prediction performance with practical deployment

requirements through efficient multimodal fusion and integration within a monitoring platform.

Unlike most previous studies that primarily focused on prediction accuracy, the proposed SENTINEL system extends the research contribution by integrating blood pressure prediction models into a complete monitoring environment. The platform supports patient monitoring, historical record tracking, alert generation, and relative supervision, providing functionality that is generally absent from existing blood pressure estimation studies.

Overall, the experimental findings are consistent with the current literature while extending previous work through a unified comparison of ECG, PPG, and RESP modalities, evaluation of multiple deep learning architectures, adaptive multimodal fusion using DGAGF, and practical deployment within a real monitoring platform.

Reference	Approach	Key Contribution	Limitation	SENTINEL Improvement
Koparr & Arslan (2024) [10]	PPG-based CNN	Deep learning BP estimation using PPG	Single modality only	Comparison of ECG, PPG, and RESP
Xu et al. (2024) [9]	ECG + PPG Fusion	Improved prediction using multimodal signals	Prediction-focused only	Added monitoring platform and history tracking
Fan et al. (2025) [11]	Transformer-based BP Estimation	Captures long-term dependencies	High computational complexity	More deployable monitoring framework
Hassan et al. (2025) [12]	Multimodal Fusion Network	ECG, PPG, RESP fusion	Research-oriented solution	Relative monitoring, alerts, and web platform
SENTINEL (Proposed)	AI-based BP Prediction + Monitoring Platform	Multimodal learning, DGAGF fusion, monitoring history, alerts, and relative supervision	Wearable integration and clinical validation remain future work	—

5.5 Limitations

Although the proposed framework achieved promising results for cuffless blood pressure estimation, several limitations should be acknowledged.

First, the study was conducted using the MIMIC-BP dataset only. While the dataset is widely used and carefully curated, evaluating the proposed models on additional public datasets would provide further evidence of their generalizability across different populations and acquisition conditions.

Second, the developed models were trained and evaluated in an offline experimental setting using pre-recorded physiological signals. Real-time validation using wearable devices and continuous physiological signal acquisition was beyond the scope of the current study.

Third, the performance of the prediction framework depends on the quality of the physiological signals. Noisy, incomplete, or improperly recorded signals may negatively affect prediction accuracy.

In addition, although the proposed multimodal fusion framework demonstrated improved performance, the computational complexity of multimodal models is generally higher than that of single-modality approaches, which may increase processing requirements in resource-constrained environments.

Finally, the current implementation focuses on blood pressure estimation and monitoring. Additional clinical validation and large-scale real-world testing would be required before deployment in healthcare settings.

Despite these limitations, the experimental results demonstrate the effectiveness of the proposed approach and provide a strong foundation for future research and practical development of cuffless blood pressure monitoring systems.

5.6 Testing (Mandatory CS Component)

Test Case	Input	Expected Outcome	Actual Outcome	Pass / Fail
Patient Registration	Valid patient information	Account created successfully	Account created successfully	Pass
Patient Login	Valid Patient ID and password	User accesses patient portal	User accesses patient portal	Pass
Relative Registration	Valid relative information	Account created successfully	Account created successfully	Pass
Relative Login	Valid credentials	User accesses relative portal	User accesses relative portal	Pass
Patient Linking	Valid Patient ID	Relative account linked to patient	Relative account linked successfully	Pass
Signal Upload	Valid ECG/PPG/RESP file	File accepted and processed	File uploaded successfully	Pass
Blood Pressure Prediction	Uploaded physiological signal	SBP, DBP, and status displayed	Prediction generated successfully	Pass
History Retrieval	Existing patient record	Previous predictions displayed	History displayed correctly	Pass
Invalid Login	Incorrect credentials	Access denied	Access denied with error message	Pass
Invalid File Upload	Unsupported file format	Upload rejected with warning	File rejected successfully	Pass

To verify the functionality and usability of the SENTINEL platform, a series of functional and graphical user interface (GUI) tests were conducted. The testing process focused on validating user authentication, patient-relative linking, physiological signal upload, blood pressure prediction, and monitoring history retrieval.

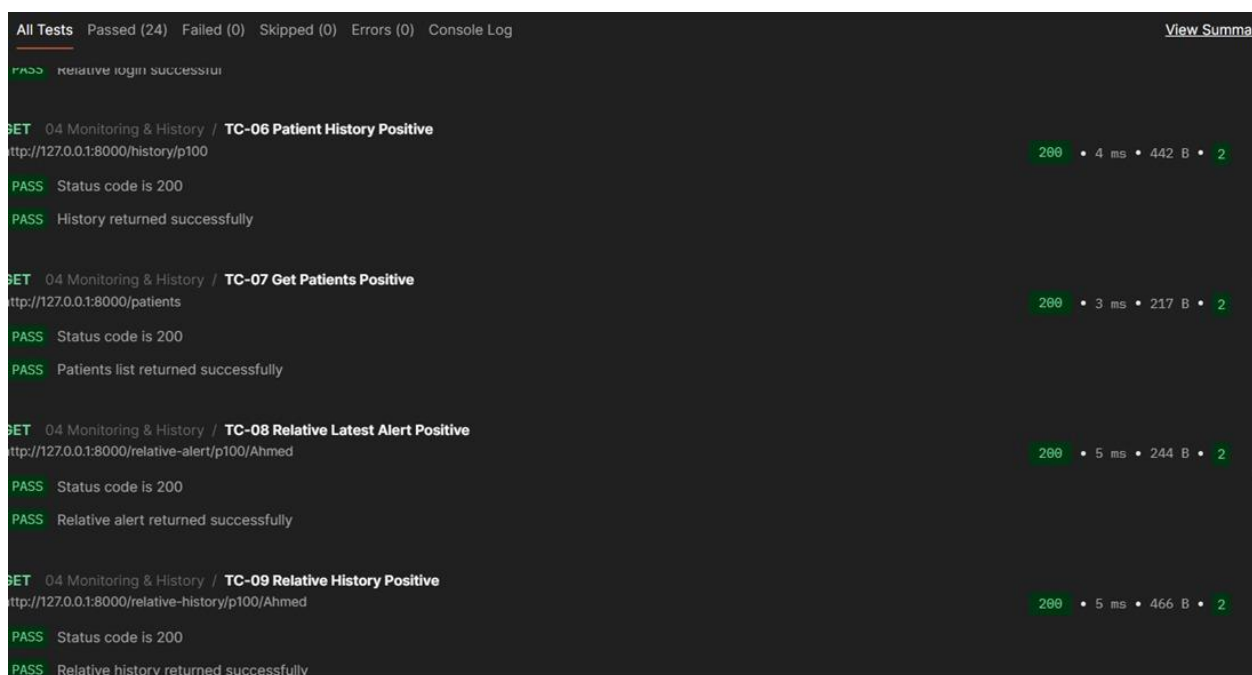
The results demonstrated that all major system functionalities operated as expected. User authentication was successfully validated for both patient and relative accounts. Signal upload and preprocessing workflows

were tested using valid physiological signal files, and the system successfully generated blood pressure predictions and health status results. In addition, monitoring history records were correctly stored and retrieved through the dashboard interface.

GUI testing was performed on the web-based platform to ensure proper navigation, responsive layout behavior, and usability across different screen sizes. Input validation mechanisms were also tested to confirm that invalid credentials and unsupported file formats were handled appropriately.

The testing results indicate that the implemented system satisfies its functional requirements and provides a reliable interface for blood pressure monitoring and prediction.

Figure: API and integration testing results showing successful execution of system functionalities.



Chapter 6: Conclusion and Future Work

6.1 Summary of Developed Software Product

SENTINEL (Smart Elderly Navigation & Telehealth Integrated Lifeline), an artificial intelligence (AI) -based health monitoring system designed to perform cuff-less blood pressure prediction and monitoring of elderly people, was proposed in this thesis. The proposed system combines deep learning-based prediction of blood pressure with a web-based health monitoring system to offer an integrated approach that unites artificial intelligence and healthcare software development.

In the realm of artificial intelligence, this research focused on three physiological modalities (ECG, PPG, and RESP) through various deep learning models such as CNN, RNN, LSTM, GRU, and BiLSTM. Through experimentations, the most suitable model for each modality was determined, and different multimodality learning strategies for two and three physiological modalities were investigated. In addition to traditional multimodality learning techniques, this research introduced Dominant-Guided Auxiliary Gated Fusion (DGAGF), an innovative approach that uses the complementary nature of physiological data for feature fusion by differentiating dominant and auxiliary modalities.

In terms of software engineering, the created platform unites React, FastAPI, SQLite, and deep learning models into one end-to-end system for monitoring patients' blood pressure. The implemented functions comprise patient and relative registration, authentication, uploading of patients' physiological signals, prediction of blood pressure, visualization of the patients' health status, history of monitoring, creation of alerts, connection to emergency contacts, and generation of reports.

The evaluation process included benchmarking of the AI models, unit testing, integration testing, API testing, black-box testing, bug tracking, and usability testing. The results show that the created platform successfully combines blood pressure prediction with healthcare monitoring capabilities.

6.2 Achievement of Requirements Matrix

Table 6.1 summarizes the achievement status of the primary objectives defined at the beginning of the project.

Project Objective	Achievement Status
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Develop cuffless blood pressure prediction models	Fully Achieved
Benchmark ECG, PPG, and RESP modalities	Fully Achieved
Evaluate multiple deep learning architectures	Fully Achieved
Investigate multimodal fusion techniques	Fully Achieved
Propose the DGAGF fusion framework	Fully Achieved
Develop the SENTINEL monitoring platform	Fully Achieved
Integrate AI models with the software platform	Fully Achieved
Implement patient and relative management	Fully Achieved
Store prediction history and alerts	Fully Achieved
Perform software testing and validation	Fully Achieved
Conduct usability evaluation	Fully Achieved

6.3 Engineering Limitations

While the SENTINEL system successfully accomplished its major goals, there are still some limitations associated with the technology. First, the created platform can be considered as a research prototype that has been tested under experimental conditions, not in real healthcare settings. Therefore, further validation will be necessary prior to the implementation.

The artificial intelligence prediction models have been trained and validated using the MIMIC-BP dataset, which is publicly available. While data splitting by subjects has been used for validation purposes, it would be beneficial to test these models on other datasets.

Moreover, the present system uses physiological signal files of .npy format for upload and does not provide any means of integrating with wearables or physiological signals in real time. Cloud-based operation, sophisticated security measures, encrypted login, and role-based access control have also been regarded as out-of-scope of the present system.

Last but not least, although the proposed DGAGF architecture showed encouraging results in the performed experiments, further clinical testing and benchmarking is needed prior to its use in actual healthcare settings.

6.4 Future Enhancements

There are several possible ways to expand upon the proposed SENTINEL solution in future research works.

From an artificial intelligence standpoint, future research might explore working with larger physiological datasets, personalizing the blood pressure estimator, using self-supervised learning, employing Transformer neural network architecture, as well as applying other multimodal fusion techniques in order to increase the accuracy of predictions and generalization ability.

From the software engineering point of view, future iterations of the solution may involve implementing real-time sensor integration, cloud computing, stream processing of physiological signals, secure authentication and role-based access control features, automated notifications and electronic healthcare information system integration.

Clinical validation of the solution on a large scale with the participation of healthcare professionals and elderly patients is another possible area of investigation in future research.

Individual Contributions

Student	Individual Contribution	Shared Contribution
Rawan Ali	Frontend development, Backend implementation, FastAPI APIs, React UI, Database integration, AI model deployment, UML diagrams, Testing documentation, Thesis writing	Dataset preparation, AI experiments, Model evaluation, DGAGF evaluation, Presentation, Final validation
Menna Hossameldin	AI model training, Dataset preprocessing, Deep learning experiments, Multimodal fusion implementation, Performance evaluation, Result analysis, Thesis writing	Dataset preparation, AI experiments, System integration, Presentation, Final validation

References and Appendices

Every reference cited in the text must appear here, and vice versa. Use the required citation style.

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